

ECE228 Final Project Report: Solar Power Generation Prediction

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Abstract

Solar photovoltaic generation is strongly affected by weather, atmospheric conditions, and solar geometry. This variability creates a forecasting problem for grid scheduling, energy storage, and short-term renewable power operation. In this project, we study solar power generation prediction as a supervised regression task using a public Kaggle dataset with meteorological variables, cloud-cover measurements, wind conditions, solar-position features, and measured generated power. We implement a model-comparison pipeline that begins with linear and regularized linear baselines, then evaluates support vector regression, random forest regression, gradient boosting regression, and neural-network regression. The best model is gradient boosting, which achieves a test MAE of 256.56 kW, RMSE of 387.06 kW, and R^2 of 0.8249. Interpretation using correlations, tree-based importance, permutation importance, and SHAP indicates that angle of incidence, azimuth, zenith, shortwave radiation, and cloud cover are the dominant drivers of predicted solar output. The results show that nonlinear tabular machine learning models can substantially improve over linear baselines while retaining physically meaningful feature behavior.

Keywords: solar power forecasting, photovoltaic generation, supervised regression, meteorological data, feature importance, machine learning

1 Introduction

Solar photovoltaic (PV) systems are increasingly used in residential, commercial, and utility-scale energy production because solar energy is renewable, widely available, and compatible with distributed generation. However, unlike conventional power plants, PV output cannot be directly dispatched by operators. The generated power changes with solar irradiance, cloud cover, air temperature, humidity, wind conditions, and the position of the sun. These environmental dependencies make solar energy integration challenging: grid operators must balance supply and demand even when renewable generation changes rapidly.

Accurate short-term solar power prediction can reduce this uncertainty. If a system predicts low PV output, operators can prepare backup generation or storage discharge in advance. If high output is expected, operators can schedule storage charging or energy delivery more efficiently. This makes solar generation forecasting a natural physical application of machine learning: the input variables have direct physical meaning, and the target variable is a continuous power measurement.

This project focuses on predicting generated solar power from meteorological and solar-position features. We use the Solar Energy Power Generation Dataset on Kaggle, which contains 4,213 observations and 21 columns: 20 raw input features and one target variable, `generated_power_kw`. The input variables include near-ground temperature, relative humidity, sea-level pressure, precipitation, snowfall, cloud cover at multiple altitudes, shortwave radiation, wind speed and direction at several heights, wind gust, angle of incidence, zenith, and azimuth.

Our contributions are fourfold. First, we build a reproducible preprocessing pipeline with train-test splitting, standardization where appropriate, and circular encoding for angular variables. Second, we compare seven regression models ranging from linear baselines to nonlinear ensemble and neural models. Third, we interpret the final model using correlation analysis, model-based feature importance, permutation importance, and SHAP. Fourth, we analyze errors across generation levels, radiation conditions, and cloud-cover regimes to identify where the prediction task remains difficult.

2 Related Work

Prior studies show that PV output forecasting benefits from nonlinear regression models because solar generation depends on interacting meteorological variables. Tripathi et al. compared multivariate regression, support vector machine regression, and Gaussian process regression for PV panel output prediction under changing environmental

conditions [1]. Their work found support vector machine regression to be effective, suggesting that nonlinear kernels can capture relationships that are difficult for linear models.

Shakhovska et al. studied solar energy forecasting with detailed meteorological data and compared several machine learning and deep learning models, including support vector machines, k-nearest neighbors, CatBoost, LSTM, and LSTM with inception layers [2]. A key contribution of their work is the emphasis on data preprocessing and feature selection. They also introduced a daylight-hour wrapper to avoid making positive power predictions during hours when solar production should be near zero. This is relevant to our dataset because solar-position variables such as zenith, azimuth, and angle of incidence encode physical information about daylight and panel orientation.

Rizk-Allah et al. proposed an explainable solar power forecasting framework based on LSTM, equilibrium optimization, and LIME explanations [3]. Their work motivates the interpretability component of our project: a strong forecasting model is more useful if it can also identify the environmental factors that drive its predictions. While their model uses a more complex optimized deep-learning architecture, our project emphasizes a compact comparison of classical and neural regression models that can be implemented and interpreted within the ECE 228 project scope.

Compared with these works, our project is intentionally focused on a smaller tabular dataset and a transparent evaluation pipeline. We do not attempt to build a large sequence model or optimize a deep architecture. Instead, we ask which common regression models work well on the available physical variables and whether the selected model agrees with solar-generation intuition. This makes the study a practical model-selection and interpretation exercise rather than a new forecasting architecture.

3 Dataset and Feature Characterization

3.1 Dataset Source

The dataset used in this project is the Kaggle *Solar Energy Power Generation Dataset* [4]. The Kaggle description frames the task as short-term PV power output forecasting for grid management, energy storage, and decision-making in energy markets. The local copy used for this project is stored as `solar.csv`. It contains 4,213 rows, no missing values, 20 raw input variables, and one continuous target variable.

3.2 Input Variables

The features can be organized into six physical groups:

Table 1: Feature groups in the solar generation dataset.

Group	Variables
Thermodynamic	temperature, humidity, pressure
Precipitation	precipitation, snowfall
Cloud cover	total, high, medium, low cloud cover
Radiation	shortwave radiation
Wind	wind speed, direction, gust at multiple levels
Solar geometry	incidence angle, zenith, azimuth

The target variable is generated power in kilowatts. In the local dataset, generated power ranges from approximately 0 kW to 3056.79 kW, with a mean of 1134.35 kW and a standard deviation of 937.96 kW. The median is 971.64 kW, and the interquartile range is from 231.70 kW to 2020.97 kW. Figure 1 shows that the dataset contains low-, medium-, and high-generation cases rather than a narrow operating range. Shortwave radiation ranges from 0 to 952.30 W/m², and total cloud cover ranges from 0 to 100, so the data includes both clear and cloudy conditions.

3.3 Physical Interpretation

Several variables have direct physical relationships with solar generation. Shortwave radiation measures incoming solar energy and should be positively correlated with power. The angle of incidence and zenith angle describe the relative position of the sun and panel; larger incidence angles generally reduce effective irradiance on the panel surface. Cloud-cover variables reduce incident radiation by scattering or blocking sunlight. Temperature can affect PV conversion efficiency because panel efficiency typically decreases at high operating temperatures, although air temperature may also correlate with sunny conditions. Humidity, precipitation, and snowfall can reduce output indirectly by changing atmospheric transmittance or blocking panels.

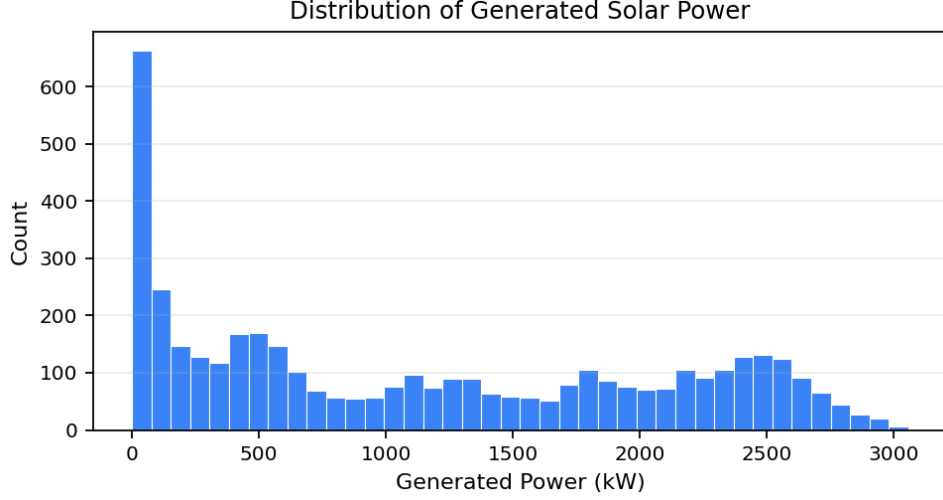


Figure 1: Distribution of generated solar power in the dataset. The target covers very low, medium, and high generation regimes, motivating both regression metrics and later grouped error analysis.

Wind features may have two competing effects. Strong wind may correlate with weather systems and cloud movement, but it may also cool panels and improve efficiency. Directional variables such as wind direction and azimuth are circular; therefore, direct numeric treatment can introduce artificial discontinuities between values near 0° and 360° . We account for this issue during preprocessing.

4 Problem Formulation

We define the task as supervised regression. For each observation i , let

$$\mathbf{x}_i \in \mathbb{R}^p$$

denote the vector of meteorological and solar-position features, and let

$$y_i \in \mathbb{R}_{\geq 0}$$

denote the generated solar power in kW. The goal is to learn a function

$$f : \mathbf{x}_i \mapsto \hat{y}_i$$

that minimizes prediction error on unseen observations.

We evaluate each model using standard regression metrics:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (1)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (2)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}. \quad (3)$$

MAE gives an interpretable average absolute error in kW, RMSE penalizes large errors more heavily, and R^2 measures explained variance relative to a mean predictor.

5 Methodology

5.1 Preprocessing

The preprocessing pipeline contains the following steps:

1. **Missing-value and range checks.** The local CSV has no missing values. We verify each column before training and inspect the distribution of generated power, radiation, cloud cover, and key solar-position variables.
2. **Train-test split.** The data is split into an 80% training set and a 20% testing set with random seed 228. The testing set is held out for final evaluation. Within the training set, cross-validation is used for model selection and hyperparameter tuning.
3. **Scaling.** Linear models, support vector regression, and neural networks are trained using standardized features. Tree-based models are trained on the original scale because they are less sensitive to feature scaling.
4. **Circular encoding.** Directional variables are encoded using sine and cosine transformations:

$$\theta_{\sin} = \sin(\pi\theta/180), \quad \theta_{\cos} = \cos(\pi\theta/180).$$

This transformation is applied to the three wind-direction variables and azimuth. The 20 raw input variables become 24 processed input features after replacing four angular columns with eight sine/cosine features.

5. **Target transformation check.** Because generated power is nonnegative and right-skewed, we inspect a $\log(1 + y)$ transformation. The final reported models use the raw kW target so that MAE and RMSE remain directly interpretable.

5.2 Models

We begin with linear regression, ridge regression, and lasso regression as baselines. Linear regression tests whether a simple additive relationship is sufficient. Ridge and lasso add L_2 and L_1 regularization, respectively, to reduce variance and test whether sparse linear structure is useful.

We then compare the baselines with nonlinear models. Support vector regression (SVR) is evaluated with an RBF kernel. Random forest regression models nonlinear feature interactions by averaging many decision trees. Gradient boosting regression builds trees sequentially to correct previous residuals. Finally, a multilayer perceptron (MLP) regression model is included as a compact neural-network baseline. Table 2 summarizes the settings used in the reported experiments; for SVR, random forest, gradient boosting, and the neural network, the listed configurations are selected through grid search on the training set.

Table 2: Model settings and selected hyperparameters used in the experiments.

Model	Selected configuration
Ridge	$\alpha = 1.0$
Lasso	$\alpha = 1.0$, max iterations = 10000
SVR	RBF kernel, $C = 100$, $\epsilon = 10$, $\gamma = \text{scale}$
Random Forest	200 trees, max depth = 20, min samples split = 2, min samples leaf = 2
Gradient Boosting	300 estimators, learning rate = 0.05, max depth = 4, subsample = 0.8
Neural Network	hidden layers (64, 32), ReLU activation, Adam optimizer, $\alpha = 0.0001$, learning rate = 0.001, early stopping

5.3 Interpretation Methods

Interpretability is important because a purely numerical forecast does not explain the physical drivers of solar generation. We use four complementary interpretation methods. First, Pearson and Spearman correlations summarize linear and rank-based relationships between each feature and generated power. Second, the gradient boosting model provides built-in feature importance values based on split quality. Third, permutation importance measures how much test RMSE increases when one feature is randomly shuffled. Fourth, SHAP values explain how individual features contribute to the final model predictions.

6 Results and Analysis

6.1 Model Comparison

Table 3 reports test-set performance for all trained models, and Figure 2 shows the same comparison visually. The nonlinear models improve over the linear baselines. Lasso is the best linear baseline with RMSE 500.90 kW and

$R^2 = 0.7067$. Gradient boosting achieves the lowest RMSE, 387.06 kW, and the highest R^2 , 0.8249. This is a 22.7% reduction in RMSE relative to the best linear baseline. Random forest has the lowest MAE, 248.77 kW, but gradient boosting is selected as the final model because it has lower RMSE and higher explained variance.

Table 3: Test-set model comparison. Lower MAE/RMSE and higher R^2 are better.

Model	MAE (kW)	RMSE (kW)	R^2
Gradient Boosting	256.56	387.06	0.8249
Random Forest	248.77	394.34	0.8182
Neural Network	291.90	424.81	0.7891
SVR	297.04	451.40	0.7618
Lasso Regression	373.37	500.90	0.7067
Ridge Regression	373.51	501.02	0.7066
Linear Regression	373.57	501.10	0.7065

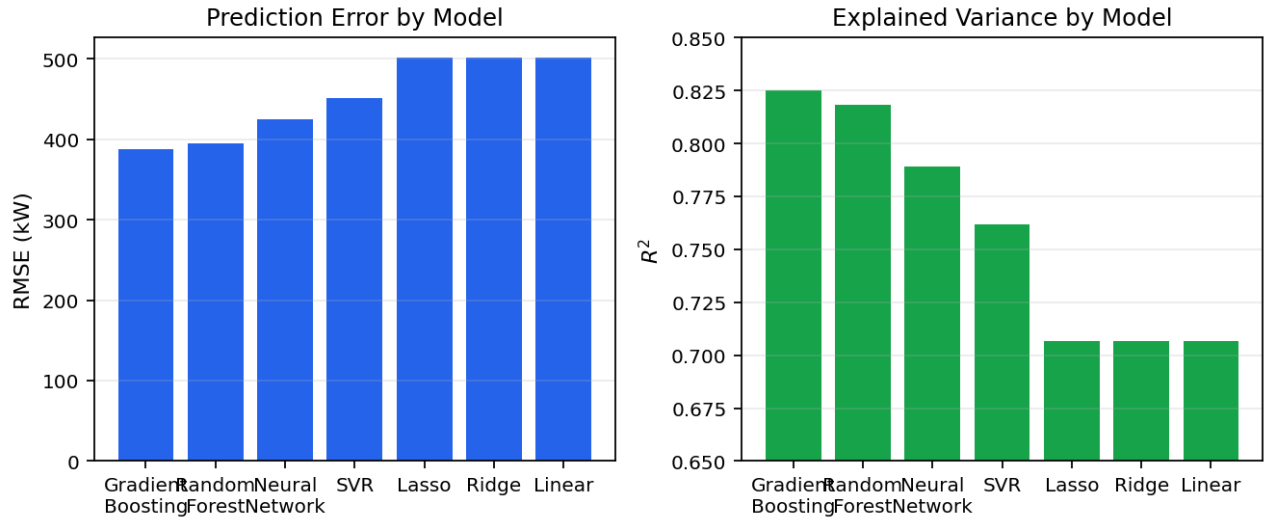


Figure 2: Model comparison on the held-out test set. The nonlinear tree-based models have lower RMSE and higher R^2 than the linear baselines, with gradient boosting giving the best overall RMSE and explained variance.

The results suggest that solar power output in this dataset is not well captured by a purely linear relationship. The largest improvement comes from tree-based nonlinear models, which can model interactions among solar geometry, cloud cover, radiation, and thermodynamic variables. The neural network improves over linear regression and SVR, but it does not outperform the tree ensembles. This is reasonable for a dataset with 4,213 observations and structured tabular features, where boosted trees often perform strongly.

6.2 Correlation Analysis

The strongest correlations are physically meaningful. Table 4 shows the top target-related features by absolute Pearson correlation. Zenith and angle of incidence are negatively correlated with power because unfavorable sun positions reduce effective irradiance. Shortwave radiation is positively correlated with power because it measures incoming solar energy. Cloud cover and humidity are negatively correlated with power, consistent with attenuation of sunlight under cloudy or humid conditions.

Correlation is only a first-order analysis. It cannot fully describe nonlinear feature interactions or conditional relationships. For example, cloud cover may matter differently at different radiation levels, and solar-position features may interact with time-of-day effects. Therefore, the correlation results are used as a physical sanity check rather than as the final interpretation.

Table 4: Top correlations with generated solar power.

Feature	Pearson	Spearman
Zenith	-0.6500	-0.6819
Angle of incidence	-0.6465	-0.6693
Shortwave radiation	0.5561	0.5539
Azimuth cosine	-0.5406	-0.5552
Relative humidity	-0.3368	-0.3029
Total cloud cover	-0.3343	-0.3115
Low cloud cover	-0.2881	-0.2661
Medium cloud cover	-0.2278	-0.2211
Temperature	0.2173	0.2100

6.3 Model-Based Feature Importance

Table 5 compares built-in gradient boosting importance with permutation importance, and Figure 3 highlights the permutation-importance ranking. Permutation importance is reported as the increase in RMSE after randomly shuffling a feature on the test set. Both methods identify angle of incidence as the most important feature. Azimuth sine, zenith, shortwave radiation, total cloud cover, and relative humidity are also important.

Table 5: Gradient boosting feature importance. Permutation importance is increase in RMSE (kW).

Feature	Built-in importance	Permutation importance
Angle of incidence	0.3603	279.38
Azimuth sine	0.1012	195.51
Zenith	0.1747	122.97
Shortwave radiation	0.0608	87.91
Total cloud cover	0.1357	69.73
Relative humidity	0.0265	64.18
Mean sea-level pressure	0.0236	36.60
Wind gust at 10 m	0.0128	16.06

The agreement between correlation, tree importance, permutation importance, and SHAP provides evidence that the model behavior is physically plausible. The final model does not appear to rely on arbitrary variables; instead, it depends most strongly on solar geometry, radiation, and atmospheric conditions. The SHAP summary plots in the project notebook provide an additional local explanation view: each feature can either increase or decrease individual predictions depending on the feature value and its interaction with the rest of the input.

6.4 Error Analysis

For the final gradient boosting model, the mean prediction error is 27.21 kW, which is small relative to the target scale. This suggests that the model does not strongly overpredict or underpredict on average. The mean absolute error is 256.56 kW, matching the model comparison table, while the largest absolute error is 1860.29 kW. The large maximum error indicates that a few cases remain difficult even though average performance is strong.

Table 6: Final-model absolute error by generated-power level.

Generation level	Count	Mean AE (kW)	Median AE (kW)	Max AE (kW)
Low	278	209.73	104.77	1860.29
Medium	287	296.86	210.51	1697.68
High	278	261.78	174.69	1423.66

The medium-generation group has the largest mean and median absolute error. This is physically plausible because medium output often corresponds to mixed conditions such as partial cloud cover, changing solar geometry, or intermediate radiation levels. Very low output cases can be easier when the sun position or radiation clearly implies weak generation, while very high output cases may be easier under consistently clear and favorable conditions.

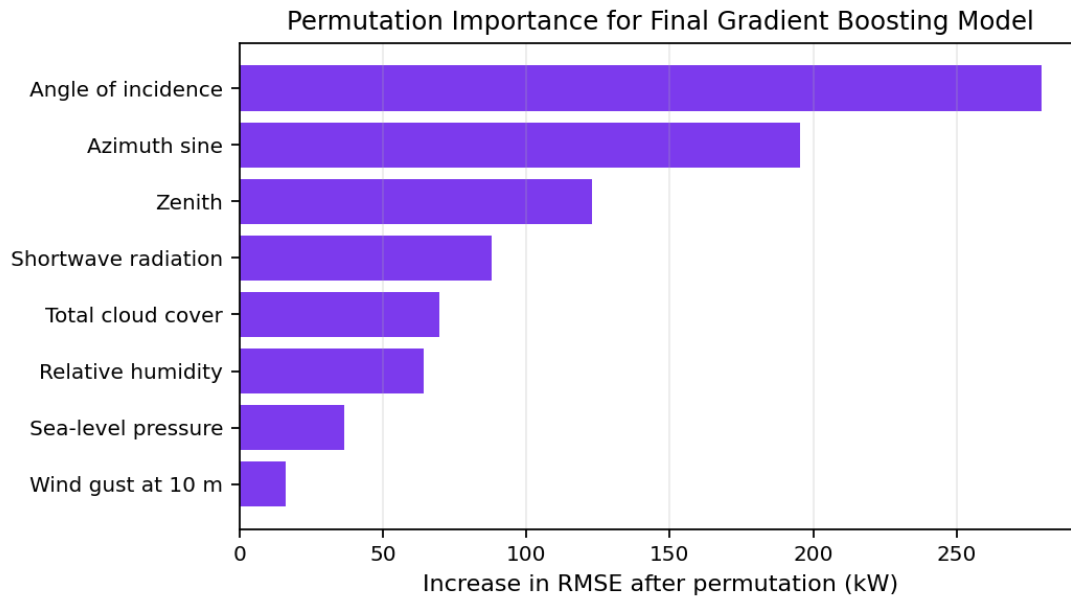


Figure 3: Permutation importance for the final gradient boosting model. Angle of incidence is the dominant feature, followed by azimuth, zenith, shortwave radiation, cloud cover, and humidity.

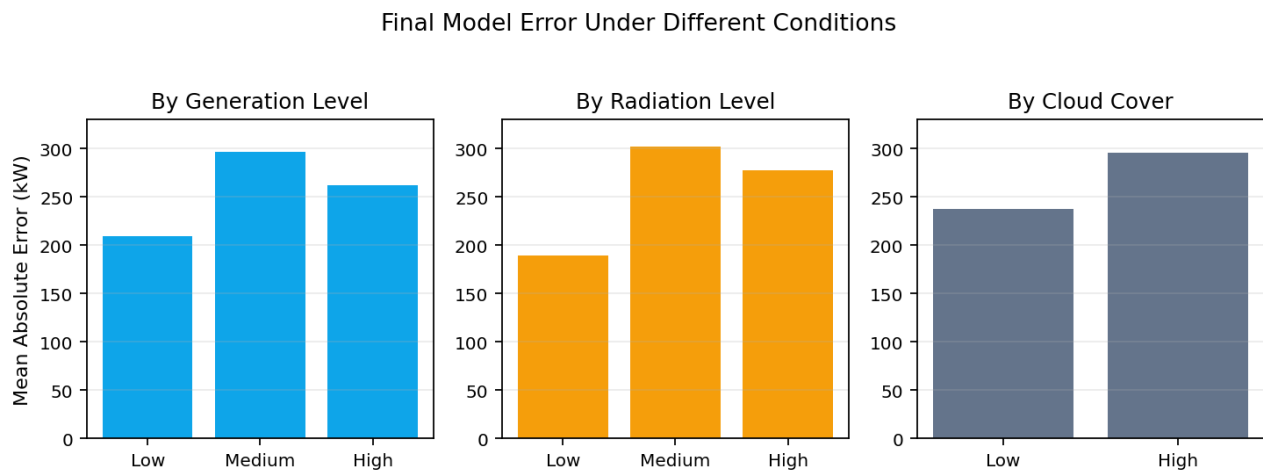


Figure 4: Mean absolute error of the final gradient boosting model across generation, radiation, and cloud-cover groups. Medium-generation, medium-radiation, and high-cloud-cover cases are harder to predict.

Table 7: Final-model absolute error by environmental condition.

Condition group	Count	Mean AE (kW)	Median AE (kW)	Max AE (kW)
Low radiation	281	189.28	125.77	984.91
Medium radiation	281	302.39	202.62	1697.68
High radiation	281	278.00	148.25	1860.29
Low cloud cover	563	237.27	130.50	1860.29
High cloud cover	280	295.34	201.59	1423.66

The condition-based analysis, summarized in Figure 4, gives the same main message. Low-radiation cases have the smallest mean absolute error, while medium-radiation cases have the largest. High cloud cover also increases error relative to low cloud cover. This indicates that the final model performs best when the environmental signal is clearer and struggles more when weather conditions make solar generation variable.

6.5 Prediction System Demonstration

After selecting gradient boosting as the final model, the notebook implements a simple prediction function. A user can provide one row of meteorological and solar-position features, the same circular encoding is applied, and the model returns a predicted power value. The output is also classified into low, medium, or high generation using the 33rd and 67th percentiles of the target distribution. This does not change the regression model, but it makes the output easier to interpret for operational use.

Representative test examples show that the prediction function works across low, medium, and high generation cases. These examples are not used as the main evidence of accuracy because they are selected illustrative cases; the full test-set metrics and error analysis provide the proper evaluation.

7 Limitations and Future Work

The main limitation is that the current study uses a random 80/20 split rather than a chronological split. If the original rows are time ordered, future work should evaluate chronological forecasting performance to better simulate real deployment. A second limitation is that the dataset is relatively small for neural models and does not include some potentially useful variables, such as explicit timestamp features, panel orientation metadata, historical lag features, or ground-truth irradiance measurements at the panel. A third limitation is that the current models predict each row independently, so they do not use temporal continuity.

Future work should test time-aware validation, add lagged weather and generation features, and compare with sequence models such as LSTM or temporal convolutional networks when enough chronological data is available. Another useful extension would be uncertainty estimation, since grid operators often need prediction intervals rather than only point forecasts. Finally, the prediction system could be wrapped into a small application that accepts weather forecasts and outputs expected generation ranges.

8 Code and Reproducibility

The project folder contains the final report source and PDF, the notebook `ECE 228 Final Project.ipynb`, the dataset `solar.csv`, the proposal PDF, and reference PDFs. To reproduce the experiments, open the notebook from the project folder so that the relative path `solar.csv` resolves correctly, then run the notebook cells in order. The core experiments require Python with `pandas`, `numpy`, `matplotlib`, and `scikit-learn`. The SHAP interpretation section additionally requires the `shap` package. All train-test splits and model searches use random seed 228 where applicable, and the main model settings are reported in Table 2.

9 Conclusion

This project studied solar PV generation prediction from meteorological, cloud-cover, wind, radiation, and solar-position variables. The final gradient boosting model achieved the best overall test performance with MAE 256.56 kW, RMSE 387.06 kW, and $R^2 = 0.8249$, substantially improving over the best linear baseline. Random forest achieved a slightly lower MAE but had worse RMSE and R^2 , so gradient boosting was selected as the final model.

The interpretation results are consistent with physical intuition. Solar geometry variables, especially angle of incidence, azimuth, and zenith, are the strongest predictors. Shortwave radiation and cloud cover are also important, while humidity and pressure provide additional information. The error analysis shows that mixed generation and medium-radiation conditions are the hardest to predict, likely because they correspond to more variable environmental states. Overall, the project demonstrates that nonlinear tabular models can provide accurate and interpretable solar generation forecasts on this dataset, while also highlighting the need for chronological validation and uncertainty-aware forecasting in future work.

10 Team Member Contributions

Table 8: Summary of team member contributions.

Team member	Contribution
Zichen Hu	Led the coding component of the project, including notebook implementation, data preprocessing, model training, model comparison, feature-importance analysis, and organization of the experimental workflow.
Zheng Dong	Contributed to the final report structure, result interpretation, figure/table organization, writing and editing, reproducibility description, and final submission preparation.
Xiaobo Li	Contributed to the project background, related-work synthesis, method and discussion writing, report review, and remaining documentation and presentation-preparation tasks.

11 SET Confirmation Screenshots

The following screenshots document the Student Evaluation of Teaching confirmation materials provided by the team for the course submission requirement.

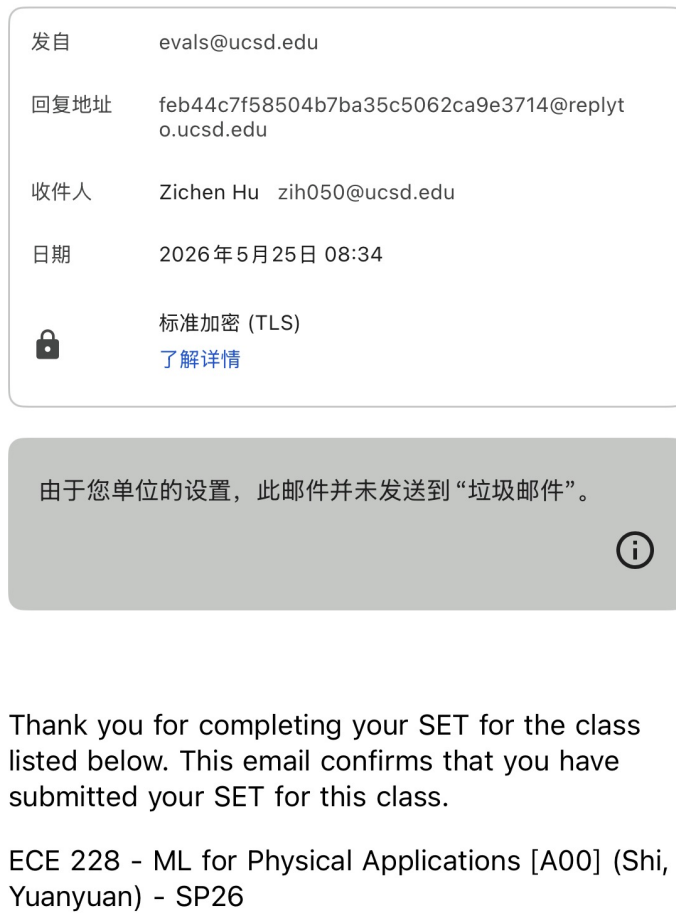


Figure 5: SET confirmation screenshot 1.

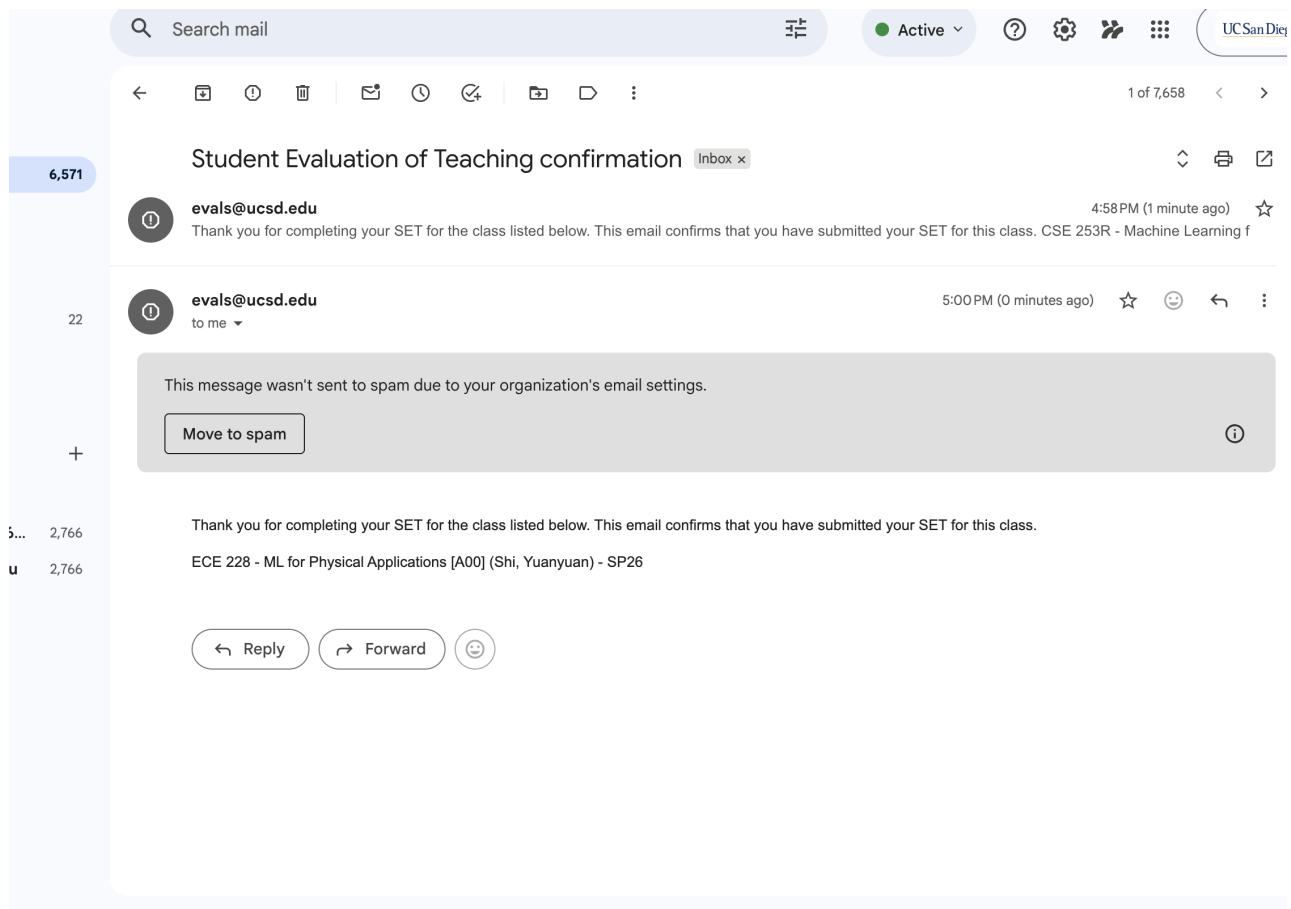


Figure 6: SET confirmation screenshot 2.

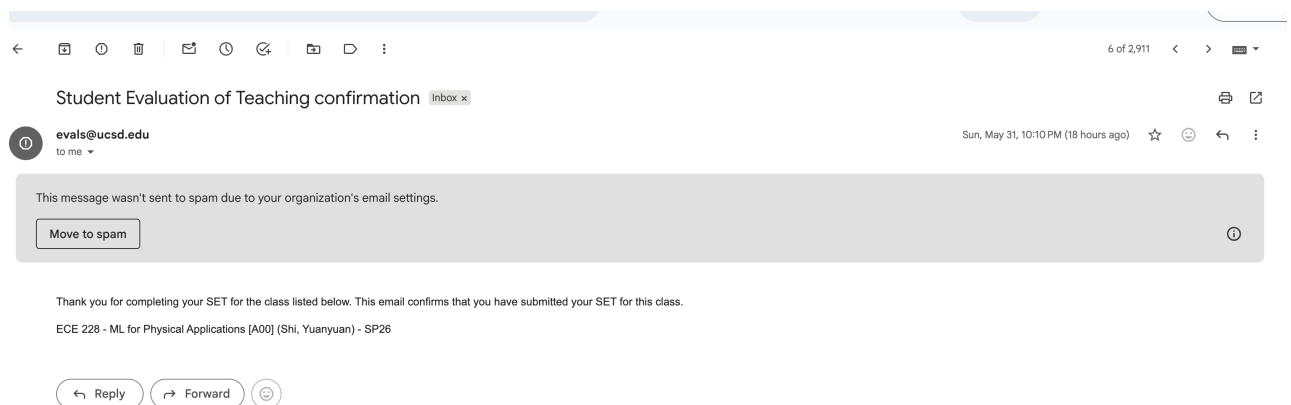


Figure 7: SET confirmation screenshot 3.

Acknowledgments

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References

- [1] A. K. Tripathi, M. Aruna, P. V. Elumalai, K. Karthik, S. A. Khan, M. Asif, and K. S. Rao, “Advancing solar PV panel power prediction: A comparative machine learning approach in fluctuating environmental conditions,” *Case Studies in Thermal Engineering*, vol. 59, 2024.
- [2] N. Shakhovska, M. Medykovskyi, O. Gurbych, M. Mamchur, and M. Melnyk, “Enhancing solar energy production forecasting using advanced machine learning and deep learning techniques: A comprehensive study on the impact of meteorological data,” *Computers, Materials and Continua*, vol. 81, no. 2, pp. 3147–3174, 2024.
- [3] R. M. Rizk-Allah, L. M. Abouelmagd, A. Darwish, V. Snasel, and A. E. Hassanien, “Explainable AI and optimized solar power generation forecasting model based on environmental conditions,” *PLOS ONE*, vol. 19, no. 10, 2024.
- [4] Stucum, “Solar Energy Power Generation Dataset,” Kaggle. Available: <https://www.kaggle.com/datasets/stucum/solar-energy-power-generation-dataset>.

Project GitHub Repository: <https://github.com/witnesszc23/ECE-228>